A B C A L C U L U S

CHAPTER 7

Differential Equations

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **7.1 – 7.8** AB

+ Extensions: **7.5** Euler's Method · **7.9** Logistic Models BC

Name: _____ Period: _____

PART I

Topics 7.1 – 7.8

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 7.1 Modeling No Calculator

A population grows at a rate proportional to the current population. Which differential equation models this situation?

(A) $\frac{dP}{dt} = k$

(B) $\frac{dP}{dt} = kP$

(C) $\frac{dP}{dt} = \frac{k}{P}$

(D) $\frac{dP}{dt} = kt$

Q2 EASY 7.2 Verifying Solutions No Calculator

Which of the following functions is a solution to $\frac{dy}{dx} = 2y$?

(A) $y = 2x$

(B) $y = x^2$

(C) $y = e^{2x}$

(D) $y = \sin(2x)$

Q3 MEDIUM 7.1 Modeling (Newton) No Calculator

A cup of coffee cools at a rate proportional to the difference between its temperature T and a room temperature of $70°$F. Which equation models this?

(A) $\dfrac{dT}{dt} = k(T + 70)$

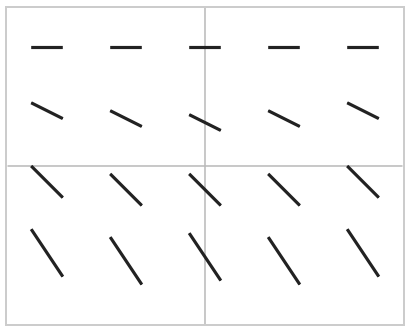
(B) $\dfrac{dT}{dt} = k(T - 70)$

(C) $\dfrac{dT}{dt} = kT$

(D) $\dfrac{dT}{dt} = 70 - kT$

Q4 MEDIUM 7.3 / 7.4 Slope Fields No Calculator

The slope field shown could represent which differential equation?



(A) $\dfrac{dy}{dx} = x$

(B) $\dfrac{dy}{dx} = y$

(C) $\dfrac{dy}{dx} = x + y$

(D) $\dfrac{dy}{dx} = xy$

Q5 MEDIUM 7.4 Reasoning No Calculator

For which differential equation are all line segments in the same horizontal row of a slope field identical?

(A) $\dfrac{dy}{dx} = x$

(B) $\dfrac{dy}{dx} = y$

(C) $\dfrac{dy}{dx} = x + y$

(D) $\dfrac{dy}{dx} = \sin x$

Q6 EASY 7.6 Separation No Calculator

If $\dfrac{dy}{dx} = \dfrac{x}{y}$, which is a general solution?

(A) $y^2 = x^2 + C$

(B) $y = x + C$

(C) $\ln|y| = \ln|x| + C$

(D) $y^2 = 2x + C$

Q7 MEDIUM 7.7 Particular Solution No Calculator

Let y be the solution to $\dfrac{dy}{dx} = 2xy$ with $y(0) = 3$. Then $y(1) =$

(A) 3

(B) $3e$

(C) $3e^2$

(D) e^3

Q8 HARD 7.7 Domain No Calculator

Let y be the solution to $\dfrac{dy}{dx} = y^2$ with $y(0) = 1$. The largest open interval containing 0 on which the solution is defined is

(A) $(-1, 1)$

(B) $(-\infty, 1)$

(C) $(-\infty, \infty)$

(D) $(0, 1)$

Q9 EASY 7.8 Exponential Growth No Calculator

A quantity Q satisfies $\dfrac{dQ}{dt} = 0.05\,Q$ and $Q(0) = 200$. Then $Q(t) =$

(A) $200 + 0.05t$

(B) $200\,e^{0.05t}$

(C) $200 \cdot 1.05^t$

(D) $\dfrac{200}{e^{0.05t}}$

Q10 MEDIUM 7.8 Decay (Half-Life) Calculator

A radioactive substance decays so that $\dfrac{dN}{dt} = -0.04\,N$, where t is in years. To the nearest year, the half-life is

(A) 14 years

(B) 17 years

(C) 20 years

(D) 25 years

Q11 MEDIUM 7.2 Verifying No Calculator

Which of the following is NOT a solution to $\dfrac{dy}{dx} = y$?

(A) $y = e^x$

(B) $y = 2e^x$

(C) $y = -e^x$

(D) $y = e^{2x}$

Q12 HARD 7.4 DE + Tangent No Calculator

Let $y = f(x)$ be the solution to $\frac{dy}{dx} = x + y$ with $f(1) = 2$. The tangent line to $y = f(x)$ at $x = 1$ is used to approximate $f(1.2)$. The approximation is

(A) 2.2

(B) 2.4

(C) 2.6

(D) 3.2

Q13 HARD 7.6 Separation No Calculator

Which of the following is a general solution to $\frac{dy}{dx} = \frac{y}{x}$?

(A) $y = Cx$

(B) $\ln|y| = \ln|x| + C$

(C) Both (A) and (B)

(D) $y = x + C$

Q14 MEDIUM 7.8 Exponential No Calculator

A bacterial culture has 1000 bacteria. After 4 hours, the culture has 4000 bacteria. If the growth rate is proportional to the current population, how many bacteria are present after 6 hours?

(A) 5000

(B) 6000

(C) 8000

(D) 16000

Q15 EASY 7.3 Slope Fields No Calculator

Consider $\frac{dy}{dx} = x - y$. The slope of the line segment in the slope field at the point $(2, 1)$ is

(A) -1

(B) 0

(C) 1

(D) 3

Q16 MEDIUM 7.6 Separable IVP No Calculator

If $\frac{dy}{dx} = \frac{x}{y^2}$ with $y(0) = 1$, then y when $x = 2$ equals

(A) $\sqrt[3]{4}$

(B) $\sqrt[3]{7}$

(C) $\sqrt{5}$

(D) 2

Q17 MEDIUM 7.7 Initial Conditions No Calculator

If y satisfies $\frac{dy}{dx} = ye^x$ with $y(0) = 1$, then $y(\ln 2) =$

(A) e

(B) $\sqrt{e}$

(C) e^2

(D) $2e$

Q18 HARD 7.4 Equilibrium No Calculator

Consider $\frac{dy}{dt} = y(2-y)$. The equilibrium solutions are

(A) $y = 0$ only

(B) $y = 2$ only

(C) $y = 0$ and $y = 2$

(D) none exist

PART II *Show All Work*

Free-Response Questions

Free-response answers require complete algebraic work: separation of variables, antiderivatives, $+C$, use of initial conditions, and correct solving for y. On the AP Exam, skipping any of these steps will cost points.

FRQ 1 EASY 7.6 / 7.7 Separation No Calculator

Consider the differential equation $\dfrac{dy}{dx} = \dfrac{x}{y}$, where $y > 0$.

(a) Find the general solution of the differential equation.

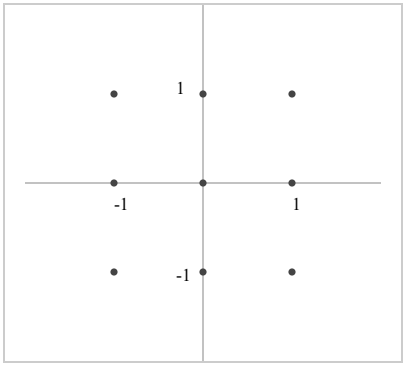
(b) Find the particular solution $y = f(x)$ satisfying the initial condition $f(0) = 2$.

(c) State the domain of the particular solution from part (b).

FRQ 2 MEDIUM 7.3 / 7.4 / 7.6 / 7.7 Slope Field + Solve No Calculator

Consider the differential equation $\dfrac{dy}{dx} = 2xy$.

(a) On the axes provided, sketch a slope field for the given DE at the nine points indicated.



(b) Find the particular solution $y = f(x)$ to the DE with $f(0) = 1$.

(c) Use your solution from part (b) to find $f(1)$.

FRQ 3 MEDIUM 7.1 / 7.7 / 7.8 Tank Leak Calculator

A tank initially contains 500 gallons of water. Water leaks out at a rate proportional to the amount remaining: $\dfrac{dW}{dt} = k\,W$, where t is in minutes. After 20 minutes, the tank contains 400 gallons.

(a) Write an expression for $W(t)$, the amount of water at time t, in terms of k.

(b) Find the value of k. Round to four decimal places.

(c) How much water, to the nearest gallon, remains in the tank after 60 minutes?

(d) At what rate, in gallons per minute, is water leaking from the tank at $t = 60$? Indicate units.

FRQ 4 HARD 7.2 / 7.6 / 7.7 Verify + Solve No Calculator

Consider the differential equation $\dfrac{dy}{dx} = \dfrac{2y}{x+1}$.

(a) Verify that $y = C(x+1)^2$ is a solution to the DE for any constant C.

(b) Find the particular solution $y = f(x)$ with initial condition $f(0) = 3$.

(c) Find the largest open interval containing $x = 0$ on which the solution from part (b) is defined, and justify your answer.

(d) Find the value of $f''(0)$ for the particular solution from part (b).

FRQ 5 HARD 7.1 / 7.4 / 7.6 / 7.7 Population No Calculator

A biologist studies a fish population $P(t)$ in a lake, where t is in years. The population is modeled by the differential equation $\frac{dP}{dt} = 0.1\,P\left(1 - \frac{P}{1000}\right)$. (You are not required to solve this differential equation.)

(a) Find $\frac{dP}{dt}$ when $P = 400$. Interpret the value in context, with correct units.

(b) For what value(s) of P is $\frac{dP}{dt} = 0$? What do these values represent?

(c) Suppose $P(0) = 400$. Use the tangent line to the graph of P at $t = 0$ to approximate $P(2)$.

(d) Use implicit differentiation to find $\frac{d^2P}{dt^2}$ in terms of P. Use this to determine whether the tangent-line approximation in part (c) is an over- or underestimate. Justify.

BC-Only Practice

BC ONLY. The following items cover Euler's method (Topic 7.5) and logistic models (Topic 7.9). AB students may skip this section. BC students should be fluent with: (i) Euler iteration $y_{n+1} = y_n + h \cdot f(x_n, y_n)$; (ii) the logistic equation $\frac{dy}{dt} = ky\left(1 - \frac{y}{K}\right)$ with carrying capacity K and inflection at $y = K/2$.

Q BC1 MEDIUM 7.5 Euler's Method BC ONLY No Calculator

Let $y = f(x)$ be the solution to $\frac{dy}{dx} = x + y$ with $f(0) = 1$. Use Euler's method with two steps of equal size $h = 0.5$ starting at $x = 0$ to approximate $f(1)$.

(A) 2.000

(B) 2.500

(C) 2.750

(D) 3.000

Q BC2 MEDIUM 7.5 Euler — Concavity BC ONLY No Calculator

Suppose the solution $y = f(x)$ to a differential equation is concave up on the interval $[a, b]$. An Euler's-method approximation of $f(b)$ starting from $f(a)$ with positive step size h will be

(A) always greater than the true value $f(b)$.

(B) always less than the true value $f(b)$.

(C) equal to the true value $f(b)$ regardless of h .

(D) indeterminate without knowing f'' .

Q BC3 EASY 7.9 Logistic — Carrying Capacity BC ONLY No Calculator

A population $P(t)$ satisfies $\frac{dP}{dt} = 0.04\,P\left(1 - \frac{P}{500}\right)$. The carrying capacity and the population value at which P is increasing fastest are, respectively,

(A) 500 and 250

(B) 250 and 500

(C) 0.04 and 250

(D) 500 and 500

FRQ BC1 HARD 7.5 Euler's Method (Tabular) BC ONLY No Calculator

Consider the differential equation $\frac{dy}{dx} = x - y$ with initial condition $y(0) = 2$.

(a) Use Euler's method with two steps of size $h = 0.5$, starting at $x = 0$, to approximate $y(1)$. Show all computations.

(b) Find $\frac{d^2y}{dx^2}$ in terms of x and y. Use this to determine whether the Euler approximation in part (a) is an over- or underestimate of $y(1)$. Justify.

(c) A second student uses Euler's method with four steps of size $h = 0.25$ and obtains a different value. Without computing it, predict whether this second approximation will be closer to or farther from the true value of $y(1)$, and explain why.

FRQ BC2 HARD 7.9 Logistic Model BC ONLY No Calculator

A wildlife biologist models a deer population $P(t)$, where t is in years, by $\dfrac{dP}{dt} = 0.2\,P\left(1 - \dfrac{P}{800}\right)$, with $P(0) = 100$.

(a) State the carrying capacity. Find $\lim_{t \to \infty} P(t)$ and justify.

(b) At what value of P is the population growing fastest? Find $\dfrac{dP}{dt}$ at that value, with units.

(c) Find $\dfrac{d^2P}{dt^2}$ in terms of P alone. Use it to identify the value of P at which the graph of $P(t)$ has an inflection point. Justify your answer using a sign analysis of $\dfrac{d^2P}{dt^2}$.

(d) Sketch a qualitative graph of $P(t)$ for $t \geq 0$. Label the inflection value and the horizontal asymptote.